Skill Forecasting and Semantic Career Pathway Optimization Ecosystem for Undergraduates

R26-IT-128

Project Proposal Report

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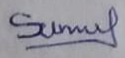
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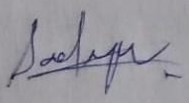
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The "Semantic Credibility Gap" That Causes Sri Lankan IT Undergraduate Students To Face Hurdles When Transitioning To The Professional Sector Due To Concealed True Technical Proficiency And Poorly Worded Resumes Data From 320 IT Undergraduates In Sri Lanka Found 62.5% Of Respondents Did Not Comprehend That There Would Be Automated Resume Screening Used During The Job Application Process. 53.1% Of Respondents Did Not Know What Skill Level They Would Be Expected To Attain Once Employed.

Academic Literature Documents A Substantial Gap Relating To The Lack Of An Existing System Which Integrates The Verification Of External Artifacts Through Analyzing GitHub Repositories To Proficiency Weighting Matching And Providing Explainable Feedback That Is Customized To The Needs Of Sri Lankan IT Undergraduates . This Research Proposes A Hybrid Proficiency Validation Framework (HPVF) Consisting Of Four Components: The Use Of A Neural Network To Extract Semantically Relevant Skills From Data By Fine-Tuning The BERT-NER Model (F1=85%) The Use Of A GitHub Repository Analyzer Based On The PyGithub Package And Abstract Syntax Tree (AST), To Cross Reference The Extracted Skills Against A GitHub Repository, With An Expected 80% Validity Rate The Use Of A Support Vector Machine (SVM) To Classify Proficiency Levels Based On Proficiency Weighting, With A Weighted Average Applied To The Degree Of Fit. Beginner=0.3, Intermediate=0.7 And Advanced=1.0) and The Creation Of A CV Recommendation Engine Providing Explainable And Actionable Feedback About CV Recommendations.

The HPVF Will Be Validated By Analyzing 300+ Sri Lankan IT Undergraduate CV's And 200 Job Postings And The Results Compared to The Traditional Keyword Matching Process Will Provide Statistically Significant Improvements.

Keywords: Semantic Credibility Gap, Verification Crisis, Hybrid Proficiency Validation Framework, GitHub Code Analysis

ACKNOWLEDGEMENT

We would like to express our appreciation for the supervisors, Ms. Sadeepa Dushanthi Kuruppu (supervisor) & Ms. Kaushika Kahatapitiya (co-supervisor), for providing us invaluable assistance, guidance and support with all phases from conception to completion.

To the academic staff of the Faculty of Computing at the Sri Lanka Institute of Information Technology (SLIIT), we thank you for your technical assistance.

In addition, we wish to thank the Head of The Department and CDAP staff for assisting us in acquiring the necessary resources to conclude our project.

We wish to extend a special thank you to all undergraduate students who completed our survey, your feedback has provided us valuable insight into addressing real-world concerns that will be used in our project.

Lastly, we must thank our families and friends for being there for us throughout this project.

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LIST OF ABBREVIATIONS

HPVF - Hybrid Proficiency Validation Framework

CV - Curriculum Vitae

NLP - Natural Language Processing

NER - Named Entity Recognition

BERT - Bidirectional Encoder Representations from Transformers

SVM - Support Vector Machine

AST- Abstract Syntax Tree

API - Application Programming Interface

ATS - Applicant Tracking System

ML - Machine Learning

TF-IDF - Term Frequency – Inverse Document Frequency

SDG - Sustainable Development Goals

ICTA - Information and Communication Technology Agency of Sri Lanka

SLIIT Sri Lanka Institute of Information Technology

GitHub - Git repository hosting service

AI - Artificial Intelligence

F1 - F1 Score (harmonic mean of precision and recall)

UI - User Interface

LIST OF APPENDICES

Appendix	Description
Appendix A	Survey Questionnaire (administered to 320 IT undergraduates)
Appendix B	System Architecture Diagram – Detailed component diagram
Appendix C	System Architecture Diagram – Detailed component diagram

1. INTRODUCTION

1.1 Background and Context

- The Real-World Problem

For IT undergraduates reaching the end of their study in Sri Lanka, entering the professional workforce after many years of dedicated study is becoming an increasingly frustrating process fraught with both uncertainty and systemic exclusion. After spending 3-4 years developing technical skills via multiple areas of study and project involvement, students often find that they are not being offered interviews because their CV does not present their technical skills in a manner that currently exists within modern recruitment systems. For example, if a student has developed a full-stack application in React.js and Node.js, their CV will only reflect "experience in web development", even though the job description clearly states that they require React.js development experience. When the automatic screening system for this job description does not match any keywords to the application it is rejected without any human review. In the absence of the candidate receiving any reason for why they were excluded from an interview the employer will never have an opportunity to discover a highly qualified candidate. This phenomenon has been termed as the "Semantic Credibility Gap" which is representative of the gap between legitimate technical competencies versus the terms used to represent those competencies.

The problem is made worse by the increasing number of AI-related resume builder programs which allow individuals to write "polished" resumes that contain all the keywords that prospective employers are looking for, regardless of whether or not they possess those skills. Students who have actually done real projects are now competing with resumes that have been built up through the use of keyword stuffing which has created a "Verification Crisis" in the job market in which true competency cannot be separated from the highly qualified candidates with resumes created through the use of keyword stuffing. Students do not understand what employers want because employers ask for "experience". however, they do not clarify if university experience counts or if there is a difference between being a beginner, intermediate or advanced level person. Students cannot see whether their applications have received any responses and thus cannot diagnose their application status; that is, they cannot determine if they lack the required experience, poorly communicate their experiences or are being eliminated from consideration due to technical requirements they have met but were not aware of.

- Statistical Evidence

Survey data from 320 IT undergraduate students reveals the depth of this confusion. 62.5% do not understand how automated Applicant Tracking Systems evaluate their CVs, submitting applications without comprehension of the filtering criteria applied to their documents. 53.1% cannot determine what proficiency level employers expect when job advertisements request specific skills, unable to calibrate their self-assessment against industry expectations. 43.8% encounter technical knowledge gaps during interviews gaps they did not know existed until evaluated, indicating that their CVs communicated proficiency they could not demonstrate. 84.4% express that a tool providing match scores and specific improvement suggestions would be useful, and 93.8% prefer integrated platforms over separate tools, indicating desire for comprehensive rather than fragmented guidance. At a national level, the Information and Communication Technology Agency (ICTA) projects growth of Sri Lanka's digital economy to USD 15 billion by 2030, requiring a skilled workforce yet each year approximately 20,000 IT graduates enter the job market while employers report

difficulty filling positions, a mismatch that impedes both individual career outcomes and national economic targets [1] .

- Limitations of Existing Solutions

The current CV writing tools for students are ineffective in preparing them for job applications. Online CV checkers provide generic advice without specific guidance on relevant terminology or applicant tracking systems. Job listings are often vague, leaving students unclear about required qualifications such as "cloud computing." College career services lack the resources to assess numerous CVs individually. AI resume builders exacerbate the issue by generating CVs filled with common keywords obscuring candidates' unique qualifications. Additionally these tools do not evaluate evidence such as GitHub projects or academic transcripts relying instead on self-reported information.

- Proposed Solution

The research aims to develop a Semantic Skill Validator and Competency Verification Framework to enhance the connection between candidates and employers. This framework includes three main components. First, it validates student's competency claims through analysis of their code repositories and associated documentation, shifting from self-reported qualifications to evidence-based assessments. Second, it assesses student's proficiency levels beginner, intermediate, or advanced based on project complexity and descriptions. Third, it offers match scores that indicate job fit by applying a weighted matching algorithm correlating proficiency with local job requirements. The framework will be validated against actual data from IT undergraduate students in Sri Lanka providing insights into the effectiveness of this approach compared to traditional keyword-based competency validation methods.

- SDG and Research Cluster Alignment

Aligning with SDG 4: Quality Education, this study allows students to see how learning in school can connect to what is needed in their field of work and SDG 8: Decent Work and Economic Growth by ensuring increased employability by validating skills through the use of evidence. The research has a connection to the Centre of Excellence for Artificial Intelligence (COEAI) which will utilize Natural Language Processing to identify skills and then apply Knowledge Management to compare the outcomes of educational programs against the job skills needed in industry. The project is also related to the Software Systems and Technologies (SST) cluster through the use of creating algorithms for proficiency classifications and to match candidates based on a weighted match approach

1.2 Literature Review

1.2.1 Existing Research Approaches

- A variety of methods for resume analysis and job matching have been investigated in recent research on AI-driven career guidance. Using KeyBERT for contextual keyword extraction and Word2Vec for synonym generation, Gangoda et al. created a skill analysis system that achieved high precision in extracting educational accomplishments (0.91) and personal information (0.94) from resumes [2].

Naharpurawala et al. went beyond conventional keyword-based applicant tracking systems by using SentenceTransformer models for semantic matching between resumes and job descriptions [3]. In an empirical comparison of distance-based algorithms, Artajaya et al. discovered that Euclidean distance with Word2Vec vectorization produced the lowest average error (0.051) when compared to Jaccard distance and Cosine similarity [4]. Through their work on academic transcript analysis and CV-job description matching, Maldeniya et al. contributed Sri Lankan context and achieved 88% accuracy with SVM for skill area prediction [5]. Scopira AI, created by Sribharathi et al., achieves 91.8% matching accuracy with explainable recommendations by combining supervised machine learning and BERT embeddings [6]. Using generative AI for real-time skill gap analysis with weighted matching factors for verified skills, Sengupta et al. presented SkillBridgeAI [7]. Kumar et al. used LinkedIn scraping to create customized job recommendations and Google Gemini AI for resume analysis [8]. Most importantly, Carter et al. created SkillScope, which provides the methodological basis for external artifact verification by analyzing GitHub source code using Abstract Syntax Trees to predict fine-grained programming skills with 91% accuracy [9].

1.2.2 Existing Products and Services

- Commercial products and services in this domain include Applicant Tracking Systems such as JobScan and Resume worded that provide keyword matching functionality, job portals like LinkedIn Jobs and TopJobs.lk offering basic profile-based recommendations, and AI resume builders that assist with formatting and keyword optimization [2] [3] [8] [9]. University career guidance services in Sri Lanka, including those at SLIIT and other institutions, provide valuable human-centric counseling but face scalability limitations [5]. However, these existing systems share critical limitations: they lack external artifact verification mechanisms to validate claimed skills against actual work products such as GitHub repositories [8]. they treat all skill mentions equally without proficiency-level differentiation [2] [4] [6] [8].they suffer from semantic rigidity that fails to recognize terminological equivalence [3] [4].they provide black-box scores without explainable feedback [6].they ignore evidence from code repositories and academic transcripts [5] [9]and they lack localization for the Sri Lankan IT graduate context [5].

1.2.3 Research Gap and Critical Evaluation

- The literature reveals a clear research gap: no existing system combines external artifact verification through GitHub analysis with proficiency-weighted matching and explainable feedback tailored to Sri Lankan IT undergraduates [2] [5] [9]. While SkillScope provides foundations for code analysis, and multiple systems offer semantic extraction and gap analysis, none integrate these capabilities into a candidate-centric validation framework that distinguishes between claimed and demonstrated proficiency [2] [3] [4]. This research addresses this gap through a Hybrid Proficiency Validation Framework that verifies skills against GitHub evidence, classifies proficiency levels based on demonstrated complexity (Beginner=0.3, Intermediate=0.7, Advanced=1.0), and generates explainable recommendations for CV improvement, validated using Sri Lankan undergraduate data [5].

1.2 Research Gap

According to the literature, existing CV optimization systems have a basic flaw in that they treat resume content as self-contained truth rather than claims that need to be verified by outside sources [2] [3] [4] [6] [8]. This leads to a "Verification Crisis" in which resumes with a lot of keywords are indistinguishable from those that demonstrate actual expertise. External artifact verification against GitHub repositories [9], proficiency-level classification based on proven evidence [2] [4] [6] [8], weighted matching that takes expertise depth into account [2] [3] [4] [6] [7], explainable feedback related to evidence gaps [6], and localized validation for Sri Lankan IT graduates [5] are five crucial capabilities that are still unmet.

Current methods are still inadequate because they are unable to differentiate between resumes that are keyword-padded and those that are authentic [2] [3] [5] [8], disregard academic transcripts and GitHub repositories as sources of evidence [5] [9], and fail to compare skill depth to job requirements 53.1% of students are unable to determine expected proficiency levels [5]. Even improvements in semantic matching [3] [6] are limited to the CV document and cannot be used to compare claims to actual work. Although SkillScope showed that code analysis can predict skills with 91% accuracy [9], candidate-side CV validation has not made use of this capability.

The Hybrid Proficiency Validation Framework used in this study fills these gaps by combining weighted matching (Beginner=0.3, Intermediate=0.7, Advanced=1.0), proficiency classification (Beginner/Intermediate/Advanced) based on demonstrated project complexity and GitHub code analysis for skill verification [9]. Additionally, the framework transforms CV optimization from keyword matching into true competency validation by offering explainable evidence-based recommendations tailored for Sri Lankan IT undergraduates [5].

Feature	Proposed System	Gangoda et al. [2]	Naharpur. et al. [3]	Artajaya et al. [4]	Maldeniya et al. [5]	Sribhar. et al. [6]	Sengupta et al. [7]	Carter et al. [8]	Kumar et al. [9]
Skill Extraction from CV	✓ BERT-NER	✓ KeyBERT	✓ Sent.Trans.	✓ Custom NER	✓ Custom NER	✓ BERT+TF-IDF	✓ Gemini	✗	✓ Gemini
GitHub Verification	✓ PyGithub+AST	✗	✗	✗	✗	✗	✗	✓ (issues)	✗
Proficiency Classification	✓ Beg/Int/Adv	✗	✗	✗	✓ grade-based	✗	✗	✓ domain	✗
Weighted Matching with Proficiency	✓ 0.3/0.7/1.0	✗	✗	✗ dist. only	✓ grade wts	✓ BERT sim.	✓ weighted	✗	✗
Explainable Feedback	✓ before/after	✗	✓ match %	✗	✓ charts	✓ confidence	✓ gap reports	✗	✓ str/weak
Localized for Sri Lanka	✓	✗	✗	✗	✓	✗	✗	✗	✗
Candidate-Centric Design	✓ student	✓ both	✓ both	✓ student	✓ student	✓ student	✓ student	✗ maint.	✓ student
External Artifact Verification	✓ GitHub	✗	✗	✗	✓ transcripts	✗	✗	✓ issues	✗

0.1

2. OBJECTIVE

2.1 Main Objective

This research's primary goal is to create, develop and validate a hybrid proficiency validation framework that tackles the "Verification Crisis" in CV screening. This will be accomplished by using transformer-based natural language processing (NLP) models to automatically extract technical skills, project descriptions and GitHub links from undergraduate IT resumes, comparing the claimed competencies with evidence from public GitHub repositories. categorizing proficiency levels into Beginner, Intermediate, or Advanced based on demonstrated project complexity and producing explainable job-fit scores with weighted recommendations specific to the Sri Lankan labor market. A dataset of more than 300 resumes and 200 job postings will be used to validate the framework which will show statistically significant improvements over conventional keyword-based and BERT-based matching techniques.

2.2 Specific Objectives

1. To create a semantic skill extraction module that uses transformer-based NLP models with at least 85% F1-score accuracy to extract technical skills, project descriptions, and GitHub links from unstructured undergraduate resumes.

2. To create a GitHub repository analyzer that can access candidate repositories, parse code structures and README files, and verify claimed skills against real project evidence with an accuracy of 80%.

3. To develop a proficiency classification model that uses demonstrated project complexity to assign skill levels (Beginner, Intermediate, Advanced), and to apply a weighted matching algorithm that computes job-fit percentages using proficiency weights (Beginner=0.3, Intermediate=0.7, Advanced=1.0) against current job requirements, attaining at least 75% agreement with expert evaluation.

4. To validate the entire framework using a dataset of 200 job postings (including local and international entry-level IT roles) and more than 300 Sri Lankan IT undergraduate CVs, showing a statistically significant improvement in differentiating candidates based on demonstrated proficiency in comparison to conventional keyword-based approaches.

3. METHODOLOGY

3.1 Introduction to Methodology

The Design Science Research (DSR) methodology used in this study is suitable for creating and assessing novel IT solutions meant to address recognized organizational issues. The six stages of the DSR methodology are problem identification and motivation, solution objectives definition, design and development, demonstration, evaluation, and communication. In order to address the "Verification Crisis" in CV screening for Sri Lankan IT undergraduates, this study uses this paradigm to develop, implement, and validate a Hybrid Proficiency Validation Framework. In this experimental study, algorithmic components are developed and compared to baseline techniques [2] [3] [4] [5].

3.2 System Components and Technical Methods

Component	Description	Key Techniques & Tools
Semantic Skill Extraction Module	Processes unstructured CVs to extract technical skills, project descriptions, and GitHub URLs.	Text extraction: PDFPlumber, python-docx; Preprocessing: NLTK (tokenization, lemmatization); NER: Fine-tuned BERT (Hugging Face Transformers) trained on 200 manually annotated CVs.
GitHub Repository Analyzer	Accesses candidate GitHub repositories to verify claimed skills against actual code evidence.	Repository access: PyGithub; README parsing; Source code analysis: linguist; AST generation: tree-sitter (Python, JavaScript, Java); Verification: threshold-based confidence scoring.
Proficiency Classification and Weighted Matching Engine	Classifies skill levels based on demonstrated project complexity and computes job-fit percentages using proficiency-weighted matching.	Classification: SVM with features from TF-IDF of project descriptions and code complexity metrics; Matching: weighted algorithm with weights (Beginner=0.3, Intermediate=0.7, Advanced=1.0); Job data.
Explainable Feedback Generator	Translates match scores and skill gaps into actionable, human-readable recommendations.	Template-based natural language generation; Provides specific suggestions, learning resources, and before/after examples.

Table 0.1

3.3 Tools and Platforms

Tool / Platform	Purpose	Justification
Python 3.9+	Primary programming language	Extensive NLP and machine learning libraries; active community support.
Hugging Face Transformers + PyTorch	BERT-based NER model development	State-of-the-art pre-trained models; fine-tuning capabilities for domain-specific skill extraction.
PDFPlumber	PDF text extraction	Preserves document layout better than OCR-based tools; reliable for text-based resumes.
PyGithub	GitHub API interaction	Comprehensive access to repository data (README, source code, commits).
NLTK / spaCy	Text preprocessing (tokenization, lemmatization, stop-word removal)	Industrial-strength NLP libraries; robust handling of unstructured text.
scikit-learn	SVM classification, evaluation metrics	Rich toolkit for machine learning; easy implementation of classification and validation.
Flask	Web application framework	Lightweight, Python-native; suitable for prototype deployment and API endpoints.
MySQL	Database management	Reliable relational database for storing user profiles, job postings, logs, and match scores.

Table 0.2

3.4 Data Collection and Ethical Considerations

Three sources of information are used:

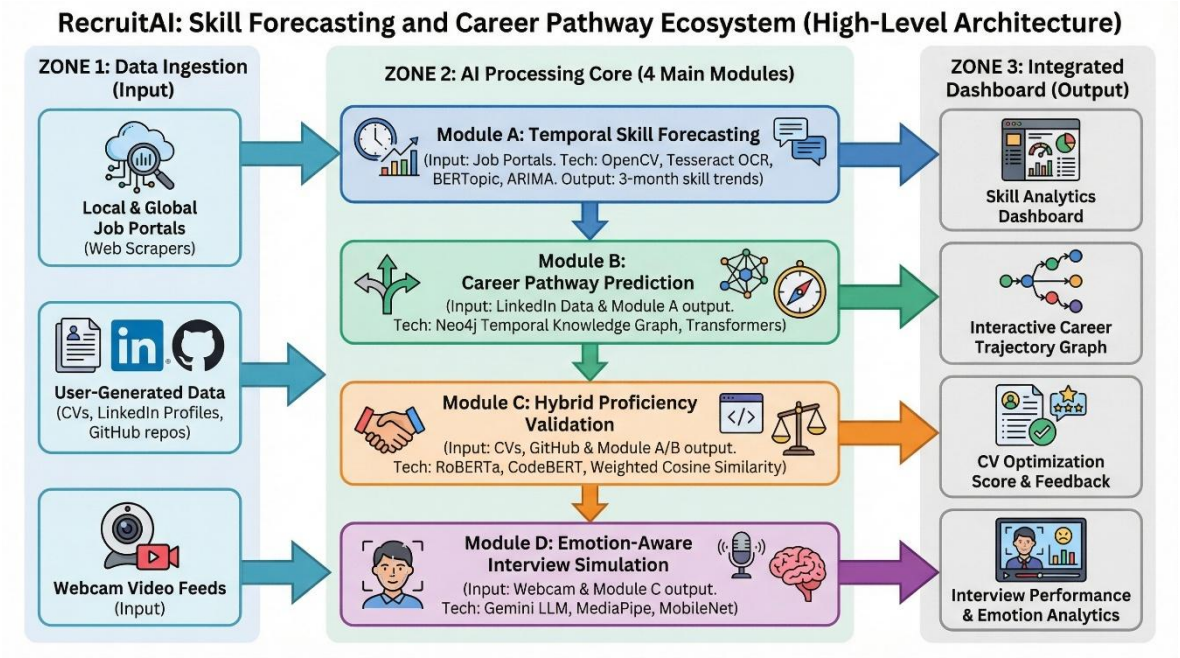
200 entry-level IT job postings from Sri Lankan portals and foreign platforms. 300 anonymized resumes from final-year IT students at Sri Lankan universities (SLIIT, University of Moratuwa, UCSC) and public GitHub repositories cited in the collected resumes. Informed consent, anonymization of personal identifiers, encrypted data storage on university servers, and adherence to GitHub's terms of service are all examples of ethical standards. The university's ethics review committee grants ethical clearance [5].

3.5 Validation Strategy

Component-level validation goals include: 75% agreement with expert evaluation for proficiency classification using Cohen's Kappa; 85% F1-score for skill extraction on 100 annotated CVs and 80% accuracy for GitHub verification against 50 manually verified CV-GitHub pairs. Using 300 resumes and 200 job postings, system-level validation evaluates the framework against conventional keyword-based matching and

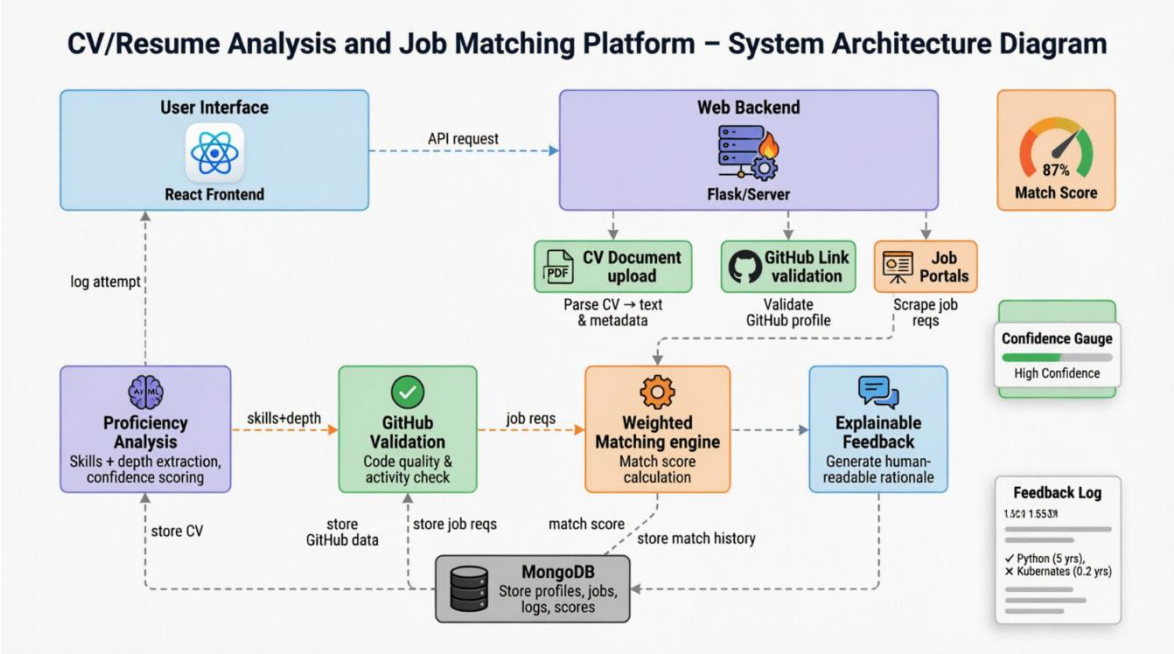
BERT-based semantic matching without proficiency weighting. Paired t-tests are used to determine statistical significance ($p < 0.05$). System Usability Scale scores, behavioral impact tracking, and uncertainty reduction (aiming for 50% via pre/post surveys) are all measured by user-centered validation with 50 students.

4. HIGH-LEVEL SYSTEM ARCHITECTURE



0.2

- **System diagram of component**



0.3

The architecture of the system is based on a client/server model with a React based front end providing the user interface for CV uploads, job searching and result visualization and a back end developed with Flask that

contains the majority of the processing logic. Upon uploading a CV, the back end parses the document, extracting text and any relevant metadata.

The back end then validates any GitHub links present within the CV. The skills extracted from the CV are then passed to the Proficiency Analysis module which calculates both the level of the skill as well as the level of confidence in the accuracy of the calculated level. While this occurs, the Weighted Matching Engine calculates a match score based upon the skill levels of the user and the job requirements scrapped from job postings on external job boards.

The Explainable Feedback module generates a human readable justification of the match. this justification is provided both as a numerical measurement of the user's confidence in their CV and as a detailed log of what skills were matched. Finally, all user profiles, job postings, logs and match scores are stored within MongoDB providing both persistence and traceability. Modularity of components of the system allows for easy expansion and maintainability.

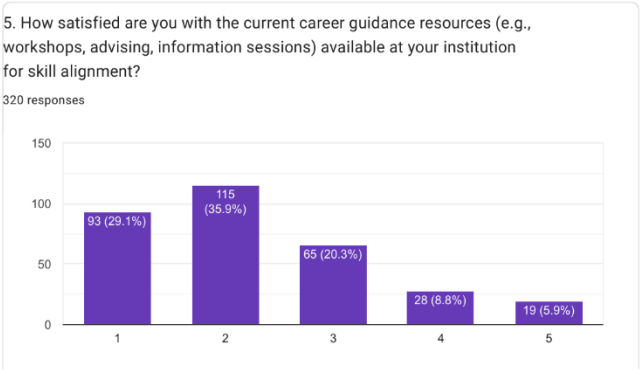
5. USER REQUIREMENTS

5.1 Stakeholder Identification

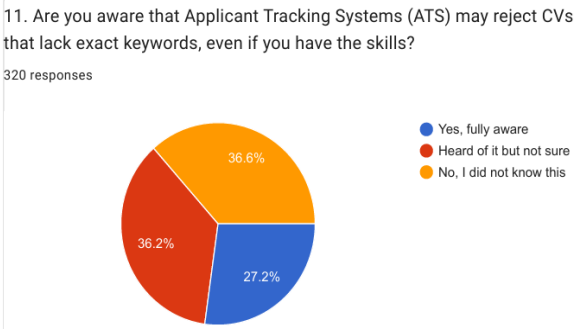
The primary stakeholders are Sri Lankan university students who are completing their final year of study in Information Technology. They will be users of the system since they will upload their resumes, be validated for skills and receive personalized job-fit scores as well as recommendations on how they can improve their chances of getting a job after graduation.

The secondary stakeholders include university career services offices and academic institutions that can utilize the platform's aggregated data about the employability of students to assist them in becoming more employable and to help align their academic programs with the needs of employers.

Although employers may derive some benefit from viewing candidate profiles as being validated, the system has been designed for students to receive transparent, evidence-based feedback to support them in building their careers not as a recruitment tool for companies.



0.5



0.4

5.2 Functional Requirements

FR-01: User Registration and Profile Management
The system shall allow students to register using email or credentials, create profiles, and securely log in. Users shall be able to upload, update, and delete CVs in PDF or DOCX format, and view version history for progress tracking.

FR-02: CV Parsing and Skill Extraction
The system shall extract text from uploaded CVs, perform preprocessing and identify technical skills, project descriptions, and GitHub URLs using a BERT-based NER model with minimum 85% accuracy. Extracted information shall be displayed for user verification and correction.

FR-03: GitHub Repository Analysis
The system shall access public GitHub repositories from CV's validate existence and ownership, parse README files, and generate Abstract Syntax Trees to analyze code patterns. Skills shall be verified by matching extracted technologies against CV claims with minimum 80% accuracy.

FR-04: Proficiency Classification
The system shall classify verified skills into Beginner, Intermediate, and Advanced levels based on demonstrated project complexity, achieving 75% agreement with expert evaluation. Users shall view their proficiency levels with supporting evidence.

FR-05: Job Matching and Recommendations
The system shall fetch real-time entry-level IT job postings via JSearch API and LinkedIn scraping, extract required skills and calculate job-fit percentages using proficiency-based weights (Beginner=0.3, Intermediate=0.7, Advanced=1.0). Top job matches shall be ranked and displayed.

FR-06: Explainable Feedback
The system shall generate personalized feedback reports identifying skill gaps, provide actionable recommendations for improving project descriptions, suggest relevant online courses, and include before/after examples. Feedback shall be presented in clear, non-technical language.

FR-07: Dashboard and Visualization
The system shall provide a user dashboard showing skill profiles, proficiency levels, match scores, and visualizations such as radar charts. Progress over time shall be displayed as users update CVs and acquire new skills.

5.3 Non-Functional Requirements

Category	Description
Performance	CV processing and job matching are optimized to provide results within a reasonable timeframe. Resource-intensive operations, such as GitHub repository analysis, are handled efficiently to ensure the system remains responsive.
Scalability	The system is designed to handle a growing number of user profiles and job postings efficiently. Computationally intensive tasks, such as GitHub analysis, are processed

	in the background to ensure smooth user experience. The architecture supports modular expansion, allowing future enhancements without disrupting core functionality.
Usability	As a web application, the interface is designed for desktop and laptop screens with a clean, intuitive layout that requires no technical background. All feedback is presented in clear, everyday language, ensuring ease of use for students.
Compatibility	The system shall support PDF and DOCX uploads. GitHub analysis shall support Python, JavaScript, and Java. The web application should function correctly on Chrome, Firefox and Edge browsers.
Ethical and Legal Compliance	All CVs used in this research are collected from public sources and fully anonymized to protect personal identity. Users provide explicit consent before their data is processed, and they retain the right to access, correct, or delete their information. Only public GitHub repositories are analyzed, strictly adhering to the platform's terms of service.

Table 0.3

6. COMMERCIALIZATION PLAN

Target market and customer persona

Target Market and Customer Persona Target Market and Customer Persona, Primary: Final-year IT undergraduates and fresh graduates in Sri Lanka preparing for job applications and technical interviews. Secondary: Academic institutions (universities, career guidance units) seeking data-driven tools to align curricula with real-time market shifts.

Value Proposition

A locally adapted AI environment offering:
Skill forecasting (3-month demand forecasts) and global-local trend analysis. Semantic CV-job matching, moving beyond keyword searching. Interview simulations, including real-time attention and emotion feedback. Visualization of career pathways based on real Sri Lankan professional data. This minimizes graduates' bench time (the need for retraining after recruitment) and helps institutions and students stay at the cutting edge of rapidly changing IT requirements.

Revenue Model

Model	Description
Premium	Basic skill analytics and career graph free; advanced features (full forecast, CV optimizer, interview simulator) via monthly subscription.

Subscription	Students pay monthly or annually for full platform access.
School Licensing	Annual B2B license for universities and career guidance units; includes institutional dashboard for curriculum planning.
Educational Partnerships	Collaborations with HR firms, and industry partners to integrate the platform into placement and employability programs.

Table 0.4

Cost Estimation

Item	Estimated Cost (LKR)
Cloud infrastructure (AWS/DO)	150,000
Database & API hosting	100,000
Developer tools & licenses	50,000
Marketing & user acquisition	200,000
Total	500,000

Table 0.5

Pricing Strategy

150,000 100,000 50,000 200,000 500,000 Student subscription: LKR 1000/month or LKR 10,000/year. · · University license: LKR 500,000/year (unlimited student access + analytics dashboard).

Freemium conversion: Target 5% of free users converting to paid after 30 days.

Competitive Advantage

Localization: Using Sri Lankan job ads and career histories instead of general global data.

Predictive not Descriptive: Using ARIMA-based 3-month forecasts to give students an edge over the competition.

Unified Ecosystem: Integrating forecasting, CV optimization, and behavioral interview simulation into one system.

Ethical Data Collection: Transparent, privacy-respecting, and complying with website terms.

Intellectual Property Considerations

- Trade secrets: Proprietary algorithms for weighted matching of skills, BERTopic-based clustering of skills, and Neo4j-based models for career pathways.
- Original IP: NLP models fine-tuned for Sri Lankan IT vocabulary and integrated feedback loops. ·
- Data ownership: All data is publicly sourced, no personal data is collected from users.
- IP Protection: Plans for future utility patent for global-local lead-lag methodology and software copyright.

7. BUDGET AND JUSTIFICATION

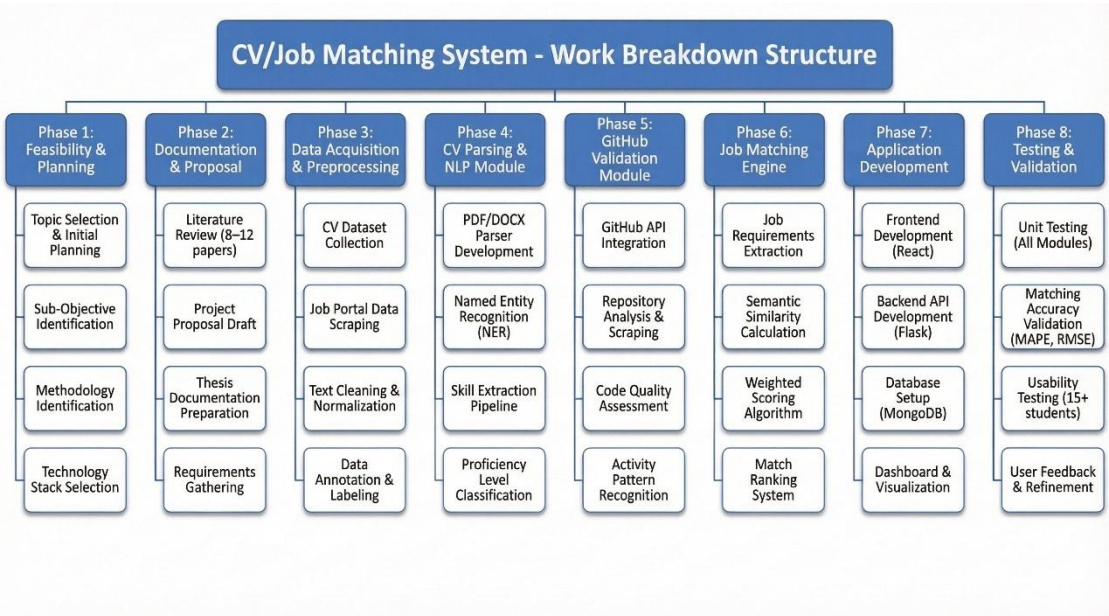
The following budget outlines the estimated costs for implementing and testing the proposed AI-driven career intelligence ecosystem. All figures are in Sri Lankan Rupees (LKR). The budget is based on a 6-month development and evaluation period with four team members

.

Category	Justification	Amount (LKR)
Hardware Costs	Webcams (x4) for real-time attention tracking in the interview simulation module (using MediaPipe / FER-2013) and an external SSD for storing scraped job posting datasets.	40,000
Software Licenses	All tools are free and open-source: Python, React, Node.js, spaCy, Tesseract OCR, MediaPipe, Neo4j Community Edition, Hugging Face Transformers, GitHub, VS Code.	0
Cloud Services	Google Cloud Run (for OCR/NLP pipeline), Gemini API (interview simulation), MongoDB Atlas, Firebase (dashboard authentication), Vercel / Render (deployment).	120,000
Human Effort Cost	Market-rate contribution of 4 members x 120 hours at LKR 250/hour each, for academic reporting purposes. No cash salaries are disbursed.	40,000
Miscellaneous	Internet costs for ethical web scraping (TopJobs.lk, Xpress.jobs, lkman.lk), API calls, and cloud access throughout the project duration.	15,000

Table 0.6

8. WORK BREAKDOWN STRUCTURE (WBS)



Phase 1 (Feasibility & Planning) will identify the subject matter and purpose of the project, as well as outline the methodology and technology to be used.

Phase 2 (Documentation & Proposal) consists of reviewing the relevant literature, writing and submitting a proposal, preparing the thesis, and gathering the necessary requirements for the project.

Phase 3 (Data Acquisition & Preprocessing) involves obtaining CVs, scraping job boards for CV feedback, removing any extraneous characters or symbols from CVs, and annotating them according to established guidelines.

Phase 4 (CV Parsing & NLP Module) includes developing the necessary software to parse the collected CVs, perform Named Entity Recognition (NER), extract identifiable skills from acquired CVs, and classify the level of proficiency.

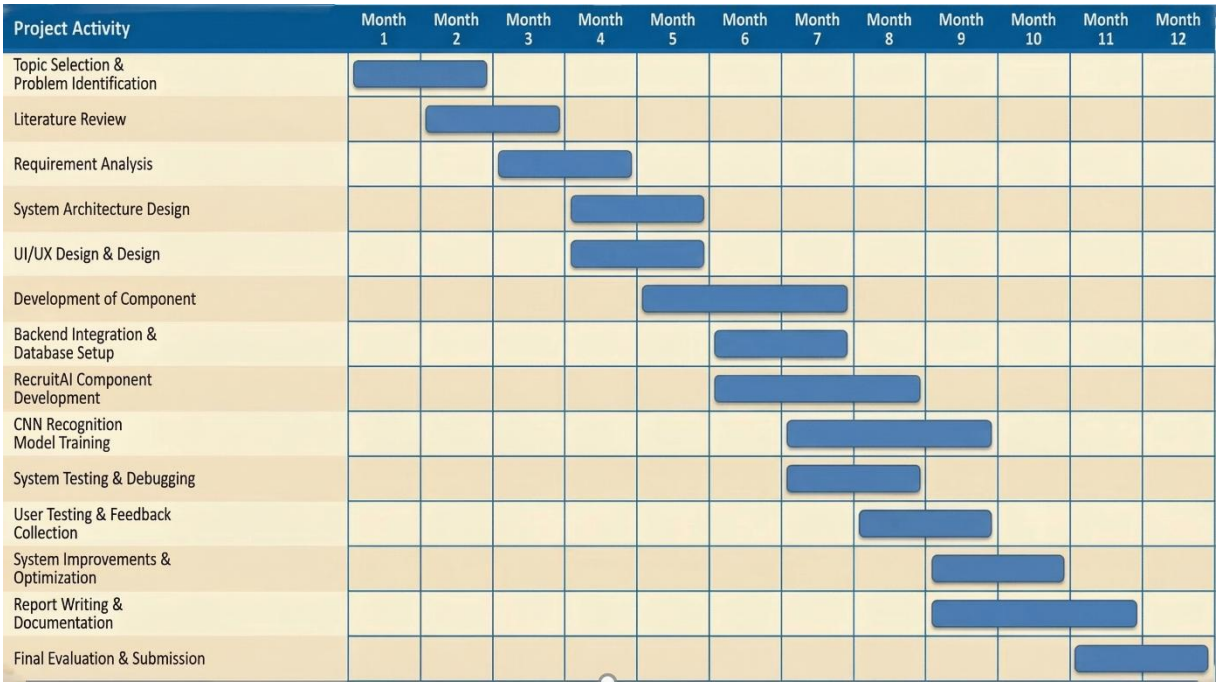
Phase 5 (GitHub Validation Module) integrates data from the GitHub API in order to perform an analysis of the activities and types of programming found within a repository.

Phase 6 (Job Matching Engine) extracts job requirement information from jobs, calculates the semantic similarity between jobs and CVs, and utilizes a weighted scoring method to assign rank.

Phase 7 (Application Development) encompasses the creation of a Frontend (web-based), Backend (API based), and Database systems for the project.

Phase 8 (Testing & Validation) completes unit testing, validating the accuracy of matches between CVs and jobs, testing usability with potential users (students), and refining the CV-Jobs matching application based on user/ tester feedback.

9. GANTT CHART



0.7

References

[1] I. a. C. T. A. (. o. S. Lanka, "https://www.icta.lk/icta-assets/uploads/2023/05/Annex-1-National-Digital-Strategy-2030.pdf," ICTA, Colombo, 2024.

[2] N. Gangoda, K. P. Yasantha, C. Sewwandi, N. Induvara, S. Thelijjagoda and N. Giguruwa, "https://ieeexplore.ieee.org/abstract/document/10756304?casa_token=wPa4mFkOJmEAAAAA:Pvp3xUlv0LAHKy5wb96zYoYgcnG3QY6Ox5lw_BA2jffcN1GWpasd8CwenwEK_5EbNUUUHh3NkgRpDw," 2024.

[3] N. J. Mohammad Naharpurawala, "Career Pilot: An AI-Powered Resume Screener and Job," https://www.researchgate.net/profile/Nachiket-Joshi-7/publication/397435622_Career_Pilot_An_AI-Powered_Resume_Screener_and_Job_Recommendation/links/6910aec54baee1659191d9c7/Career-Pilot-An-AI-Powered-Resume-Screener-and-Job-Recommendation.pdf, 2025.

[4] H. Artajaya, Julieta, J. Giancarlos, J. V. Moniaga and A. Chowanda, "Job Recommendation System based on Resume using Natural Language Processing and Distance-based Algorithm," IEEE, Bandung, Indonesia, 2024.

[5] m. m. M. D. S. S. Z. A. K. Lakmini abegunawardana, "Optimizing Candidate Selection in it Industry: An Approach for Cv-Job Description Matching and Academic Transcript Analysis," https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&q=++.+Optimizing+Candidate+Selection+in+IT+industry%3A+CV-Job+Description+Matching+and+Academic+Transcript+Analysis.+SLIIT.+%E2%80%A2++&btnG=, 2023.

[6] S. B. S. M. S. D. S. K. B Sribharathi, "Scopira: An AI-Driven Career Guidance System Using Resume Parsing, Skill Gap Analysis, and Intelligent Job Matching," https://ieeexplore.ieee.org/abstract/document/11346170?casa_token=GxwCxlg4peUAAAAA:yCS4U27k3d0w9igxlhMZ-F0V7IMVOKKKNEUSn2ZY1lYu94GMVFb1XceMzLEyLs6NyxvwQVIc8s3sRA, 2025.

[7] S. P. S. S. V. N. A. S. S Sengupta, "Real-Time AI-Based Skill Gap Analysis and Adaptive Career Guidance Using a Generative AI Framework for the Modern Job Market," https://ieeexplore.ieee.org/abstract/document/11280793?casa_token=-lXfWbR1YVwAAAAA:XnwsoiiCAe4fOCinEoMTc9YvHLgKOZdHTcHnnHhw5kEBx6V2gdlRyOqAdq42zt4IsPspBja82SGMxw, 2025.

[8] P. S. C. R. KS Kumar, "AI for career growth: advanced resume analysis and LinkedIn scraping for personalized job recommendations," https://ieeexplore.ieee.org/abstract/document/10760972?casa_token=sRZMPNVhJPoAAAAA:vaXDmG2xi-o8_wJTjWntagvcGUPhtrO0mMqw8eYu2M8alTWXgfOD9113-lryeaYdgei76uycRU9n6Q, 2024.

[9] B. C. Carter, J. R. Contreras, C. A. L. Villegas, P. Acharya, J. Utzerath and A. O. Farner, "SkillScope: A Tool to Predict Fine-Grained Skills Needed to Solve Issues on GitHub," https://ieeexplore.ieee.org/abstract/document/11029433?casa_token=lsGtbMkLm8wAAAAA:ynKIdInk0UKgE6Z6m1FD2LWbkacug-khT9tYosElrkzzyx5AYmsJbrfVxpU34Pd7uKAtvmQo6O28wg, 2025.

[10] I. Khaouja, I. Kassou and M. Ghogho, "A Survey on Skill Identification From Online Job Ads," https://ieeexplore.ieee.org/abstract/document/9517309, 2021.

[11] A. A. N. M. L. A. M Papoutsoglou, "Extracting knowledge from on-line sources for software engineering labor market: A mapping study," https://ieeexplore.ieee.org/abstract/document/8884193.

10.APPENDICES

